



Research Paper

Thermal and flow analysis of a refrigerated truck for cold chains under various conditions in practical use

Hyun Sung Hwang^{a,1}, Youngho Rhee^{b,1}, Dongchan Lee^{a,*}

^a Department of Mechanical and Information Engineering, University of Seoul, Seoulsiripdae-ro 163, Dongdaemun-gu, Seoul 02504, Republic of Korea

^b Advanced logistics Systems Lab, Korea Railroad Research Institute, Cheoldo Bangmulgwan-ro 176, Uiwang-si, Gyeonggi-do 16105, Republic of Korea

ARTICLE INFO

Keywords:

Refrigerated truck

Cold chain

Thermal flow analysis

Infiltration heat

Cargo loading

ABSTRACT

Heat infiltration from frequent door opening of refrigerated trucks during delivery for cold chains has a significant impact on food quality and energy consumption. This study investigates the flow and heat transfer characteristics inside a 15 m³ refrigerated truck. Detailed analysis was conducted during the transient state after opening the truck door under several different practical conditions. The numerical method was validated using experimental data and showed an acceptable agreement. Initially, the flow inside a refrigerated truck body with no load and the fan off was investigated after opening two doors. Subsequently, the effects of the number of doors opened, fan operation, and amount of cargo loading were analyzed. When the truck was empty and the fans were operating with all doors opened, the forced flow region was located at the top and the natural convection region was located below. The temperature and infiltration heat increased significantly after the outside air was drawn in by the fans. When the fans were turned off, natural convection was dominant, and a rapid increase in temperature occurred when the inflow of outside air increased. When only one door was closed, the circulation of cold air was maintained, suppressing air inflow and rapidly increasing the air temperature. The rate of temperature increase was approximately 23 % slower for the half-loaded case than for the no-load case. The results of this study can contribute to reducing energy consumption by optimizing the operating strategy and geometry that can minimize the heat loss after opening the doors.

1. Introduction

Refrigeration of perishable food products during transportation is essential to ensure the food's quality and safety [1]. Recently, the cold-chain market has grown significantly to meet the increasing demand for fresh food [23]. However, increased refrigerated transportation also causes significant environmental problems because it requires tremendous amounts of energy and emits massive amounts of greenhouse gases. The refrigerated transport sector accounts for approximately 15 % of the global fossil fuel energy use [4]. Furthermore, food transportation, including driving energy and refrigeration, is estimated to be responsible for approximately 2 % of total greenhouse gas emissions [5]. As numerous climate and energy regulations are being established, reducing [6] energy consumption in refrigerated transportation has become an important issue.

Frequent door opening during delivery in refrigerated trucks has a significant impact on food quality and energy consumption. Significant

amount of heat and moisture infiltration due to the inflow of atmospheric air are inevitable when doors are opened [7]. This can cause food spoilage, as the inflowing air disturbs the even temperature during refrigerated storage [8]. Furthermore, heat infiltration through open doorways results in a substantial increase in the cooling load. Existing studies have reported that warm-air infiltration can account for approximately 34 % of the total refrigeration load of refrigerated trucks in the existing studies [910]. Therefore, thermal and flow analyses of refrigerated trucks during door opening are crucial [11].

Several studies have been conducted on heat and mass transfer inside a refrigerated truck when the truck doors are opened. Sarabia Escriva et al. [12] suggested a mathematical model to estimate the additional heat due to outside air infiltration and studied the effect of inflatable shelter on energy saving; however, intensive transient study was not conducted. De Micheaux et al. [13] conducted an experimental and computational study on the infiltration heat load during the opening of truck doors. The mechanism of the fluid flow inside the truck after opening the doors was analyzed in this study. They reported that the

* Corresponding author.

E-mail address: dclee@uos.ac.kr (D. Lee).

¹ These two authors have the same contribution to this study as co-first authors.

Nomenclature		Greek symbols	
c	specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	Δ	difference (—)
g	gravitational acceleration (m/s^{-2} (— —))	ε	energy dissipation rate (m^2/s^{-3} (— —))
h_{tot}	specific total enthalpy (J kg^{-1})	θ	grid-dependency index (%)
k	thermal conductivity (W/m K^{-1})	μ	dynamic viscosity (Pa s)
m	mass (kg)	μ_{eff}	effective viscosity (Pa s)
p	Pressure (Pa)	ρ	density (kg m^{-3})
p'	modified pressure (Pa)	σ	standard deviation (—)
P_k	shear production of turbulence ($\text{kg m}^{-1} \text{s}^{-3}$)	τ	shear stress (Pa)
Q	infiltration heat load(J)	Subscripts	
T	temperature (K)	<i>air</i>	air domain
t	time (s)	<i>ini</i>	initial state
V	velocity (m/s)	<i>pre</i>	present state
		<i>t</i>	transfer

buoyancy-driven flow owing to the density difference between the inside and outside air led the heat load reach its peak at the beginning, followed by heat exchange between the outside air and the cold walls of the container. Taher et al. [14] performed thermal and flow analyses in an 8.2 m^3 refrigerated truck with different door opening and loading conditions using different Unsteady Reynolds-averaged Navier–Stokes (URANS) unsteady turbulence models (standard k- ε , standard k- ω , shear stress transport [SST] k- ω). By comparing simulation results from different turbulence models, they figured out which model shows better agreement with the experimental results. They determined that the root-mean-square error using the SST k- ω model showed the best agreement among URANS turbulence models. Numerous studies suggested solutions for heat infiltration. Moerman et al. [15] studied the air infiltration behavior of several types of doors in truck trailers. They suggested desirable door operating methods for each types of doors that can minimize infiltration. Several studies studied the effect of using an air curtain on preventing cold air leakage and analyzed its contribution to energy saving. Tso et al. [16] experimentally investigated air infiltration through doorways inside a 7.2 m^3 refrigerated truck when an air curtain was installed across the opening door. They reported that using an air curtain can reduce the energy consumption by up to 40 % and 11 % compared to cases without an air curtain and with a plastic strip curtain, respectively. Rai et al. [17] numerically analyzed the airflow behavior and temperature changes in a 44 m^3 refrigerated truck during different door opening periods to investigate the energy-saving effect of adding an air curtain. They concluded that the forced convection created by the air curtain blocked the natural convection and slowed the temperature increase in the refrigerated truck body. They also found that an air curtain

with optimum air velocity of 3.1 m/s can reduce the energy consumption up to 48 %. Some studies investigated the effect of using phase change materials in reducing the additional energy for cooling by absorbing infiltrated heat. Taher et al. [18] investigated the thermal performance of the door opening and closing processes in a refrigerated truck using three different phase change materials. They reported that using RT3 HC1, RT5, and RT8 HC significantly reduced the cooling energy required to open the truck door by 61, 57, and 35 %, respectively. Jeong et al. [19,20] conducted a numerical simulation of heat transfer during the door opening period inside a 33 m^3 refrigerated truck trailer with cold phase-change material plates. They concluded that the cargo eliminates the recirculation of the cold air inside and promotes the infiltration of atmospheric air, increasing the maximum air temperature compared to the case without the cargo. However, above studies usually focus on the overall effect after adding additional installments, not analyzing fluid flow and heat transfer during the temperature increase of truck body. Furthermore, intensive analysis investigating the effects of several variables in practical operations, such as fan operation, door opening, and cargo loading, has not been yet reported in open literature.

The main objective of this study was to analyze the flow and heat transfer characteristics of a 15 m^3 refrigerated truck after opening the doors for 40 s. Case studies were conducted under various conditions of practical use, including fan operation, door opening, and cargo loading. The numerical simulation was validated experimentally using measured data, and the predicted data showed good agreement with the experimental data. Table 1 summarizes the results of previous studies and the present study. Although previous studies have focused on the improvement of heat transfer by using additional devices such as air

Table 1
Summary of the previous studies and present study.

Author	Conditions			Key Findings
	Fan operation	Door opening (location)	Cargo loading	
Lafaye de Micheaux et al. [13]	X	1/3, 2/3, full open (rear)	0 %	Analysis of infiltration dynamics and heat transfer inside the truck
Taher et al. [14]	X	Half, full open (rear)	0 %, 25 %, 50 %, 75 %	Comparison of using different turbulent models, evaluation of the energy saving effect of using air curtain
Tso et al. [16]	O, X	Full open (side)	0 %, 25 %	Evaluation of the energy saving effect of using plastic strip curtain and air curtain
Rai et al. [17]	X	Full open (rear)	0 %, 100 %	Evaluation of the energy saving effect of using air curtain and finding the optimum air velocity
Taher et al. [18]	X	Full open (rear)	0 %	Comparison of the cooling performance of using three different phase change materials
Jeong et al. [19,20]	O	Full open (rear)	0 %, 100 %	Evaluation of the heat transfer improvement of using phase change material plates
Present study	O, X	Half, full open (rear)	0 %, 50 %, 100 %	Analysis of infiltration dynamics and heat transfer inside the truck. Effect of fan operation, door opening and cargo loading. Focus on air flow and heat transfer behavior during the transient state.

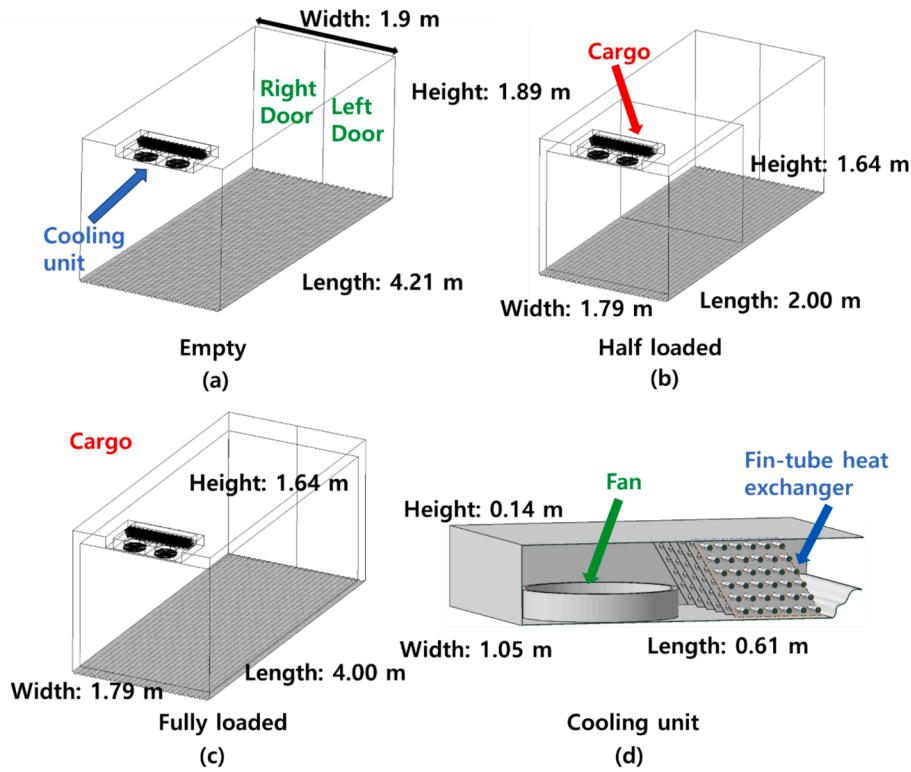


Fig. 1. Schematic views of the simulation model of (a) empty truck body, (b) half loaded, (c) fully loaded, and (d) detailed view of the cooling unit.

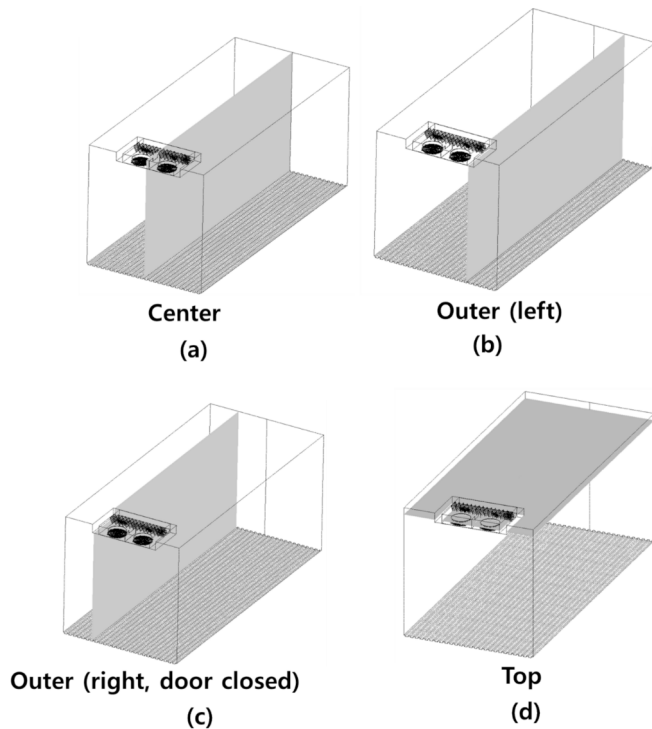


Fig. 2. Cross-sectional planes of (a) the center, (b) left side, (c) right side, and (d) top side of the truck.

curtains and phase-change materials, most of them emphasize the overall difference rather than intense analysis on each time step in the transient state. This study focuses on the detailed characteristics of the infiltration of outside air, as well as the effects of several variables in practical operation during the transient state after opening the doors. Air

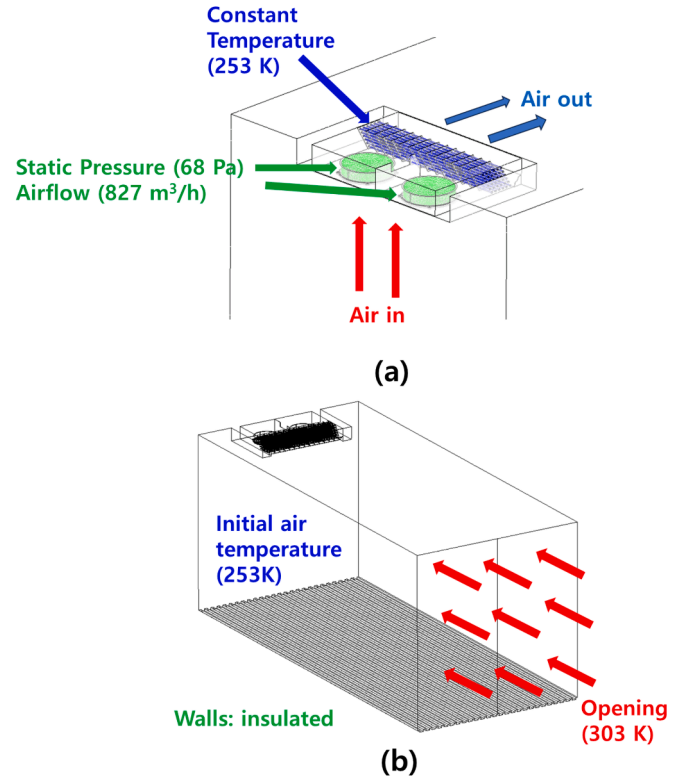


Fig. 3. Boundary conditions of the (a) cooling unit and the (b) truck body during the transient simulation.

flow and heat transfer behavior in several locations at several time steps during the transient state was analyzed. The results of this investigation can be used to determine the optimal truck operating conditions that can

Table 2
Summary of boundary conditions.

Boundary region	Boundary condition type
Fin-tube heat exchanger	Constant temperature (253 K)
Walls (truck, cargo)	Adiabatic
Outdoor air	Constant temperature (303 K)
Fan airflow	Constant flowrate (827 m ³ /h)

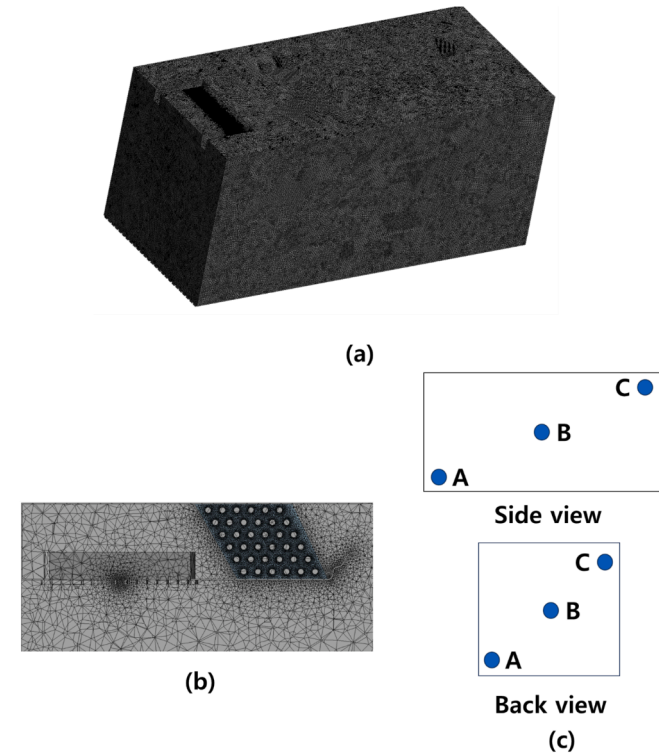


Fig. 4. (a) Schematic view of the generated mesh, (b) detailed view of the section near the cooling unit, and (c) the measurement locations for the grid dependency test.

minimize energy loss based on an understanding of heat and mass transfer after door opening under several different conditions.

2. Simulation model

2.1. Simulation domain

In the present study, the air inside a 2.5-ton refrigerated truck body was used in the simulation. Transient thermofluid simulations were

conducted using ANSYS CFX 2021 R2 with total simulation time of 40 s and time interval of 0.4 s. In the case of outdoor unloading, a 2.5-ton refrigerated truck is generally used for convenience store logistics in South Korea, therefore, unloading takes little time, mostly less than 40 s, at one destination. Fig. 1(a)–(c) show schematic views of the simulation model. As shown in Fig. 1(a), the interior space of an empty refrigerated truck body has a box-like structure with two doors and a cooling unit installed on top of the front side of the truck body. Fig. 1(b) and 1(c) show the simulation model when the truck is half-loaded and fully loaded, respectively. The cargo was a cuboid and loaded inside the truck. Fig. 1(d) shows a detailed view of the cooling unit. The cooling unit consisted of two fans and a fin-tube heat exchanger, which served as the evaporator for the refrigeration cycle.

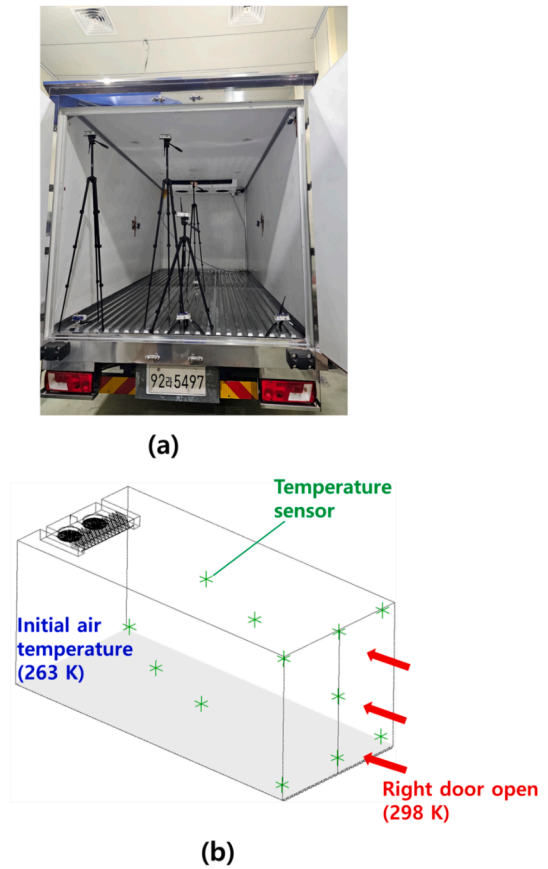


Fig. 5. (a) Photograph of the experiment equipment and the (b) schematic of the experiment settings.

Table 3
Grid dependency test results.

Number of grids (Million)	Point	2.4 s				4.8 s				7.2 s				Remark
		T (K)	θ (%)	V (m/s)	θ (%)	T (K)	θ (%)	V (m/s)	θ (%)	T (K)	θ (%)	V (m/s)	θ (%)	
19.93	A	253.23	—	0.36	—	253.37	—	0.35	—	253.43	—	0.22	—	Accept
	B	253.01	—	0.65	—	254.41	—	0.51	—	286.43	—	0.28	—	
	C	266.21	—	1.52	—	257.18	—	1.10	—	255.83	—	1.75	—	
	Avg	255.79	—	0.84	—	258.28	—	0.65	—	279.62	—	0.75	—	
17.23	A	253.14	0.04	0.35	4.35	253.30	0.03	0.34	4.12	253.30	0.05	0.20	9.50	
	B	253.01	0.00	0.61	6.72	258.99	1.80	0.48	5.42	281.92	1.58	0.26	8.85	
	C	258.55	2.88	1.50	1.33	257.26	0.03	1.00	10.22	262.28	2.52	1.70	2.65	
	Avg	255.10	0.27	0.82	3.10	258.60	0.12	0.61	7.81	279.16	0.17	0.72	4.03	
14.39	A	253.03	0.04	0.28	21.48	253.05	0.10	0.30	13.33	258.71	2.14	0.26	22.48	
	B	253.04	0.01	0.40	52.12	270.82	4.57	0.40	20.00	277.36	1.62	0.35	25.71	
	C	265.46	2.67	2.12	29.25	255.51	0.68	0.95	5.05	253.56	3.33	1.95	12.82	
	Avg	253.79	0.51	0.94	12.48	267.83	3.57	0.50	21.20	279.58	0.15	0.85	15.56	

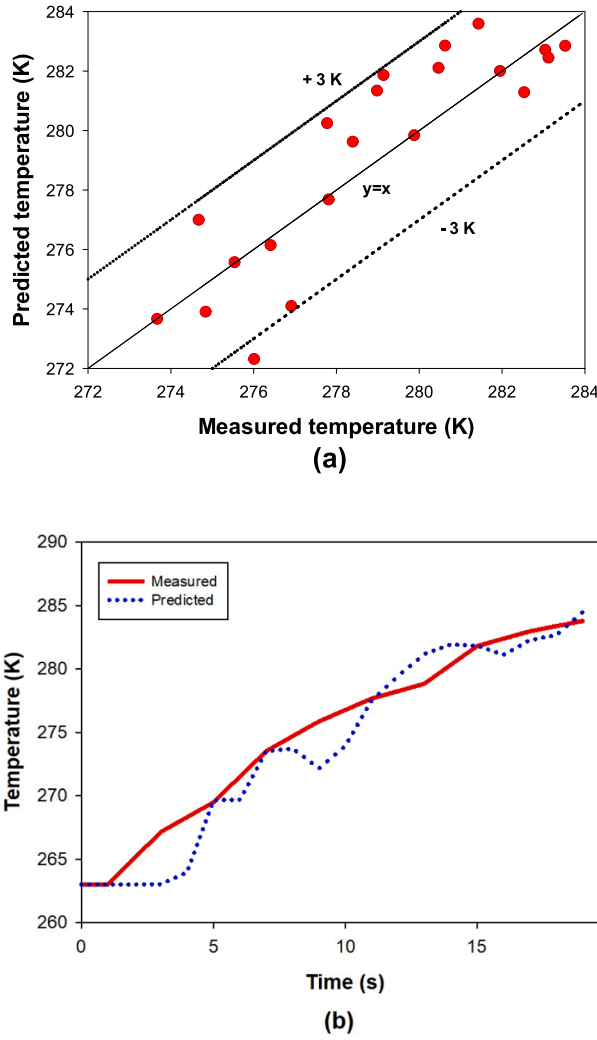


Fig. 6. Comparison between the measured and predicted temperatures.

Fig. 2 shows the cross-sectional planes used in the simulations. Fig. 2(a) shows the center of the simulation domain, which was highly influenced by the fan. Fig. 2(b) and 2(c) show the left and right sides of the simulation domain, respectively. While the two locations were less influenced by the fan than the center, the right door was closed when only one door was opened. Fig. 2(d) shows the top side of the simulation domain.

2.2. Thermofluid model

In this study, forced convection due to fans in the cooling unit and natural convection due to the opening of doors were predominant in the air domain. The governing equations in the air domain are expressed in Eqs. (1)–(3).

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho V_i}{\partial x_i} = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial \rho V_i}{\partial t} + \frac{\partial}{\partial x_i} (\rho V_i V_j) = -\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu_{eff} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \right] \quad (2)$$

Energy equation:

$$\frac{\partial}{\partial t} (\rho h_{tot}) - \frac{\partial p}{\partial t} + \frac{\partial}{\partial x_i} (\rho V_i h_{tot}) = \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} \right) + \frac{\partial}{\partial x_i} (V_i \tau) \quad (3)$$

Because the air inside the cooling unit has a Reynolds number above 2×10^5 , which is much higher than the Reynolds number required to achieve fully turbulent internal flow conditions, the κ – ϵ turbulence model was used in the simulation domain. To check the inaccuracy between turbulence models, simulation results were compared with the results using the SST k – ω turbulence model, showing trivial difference. The transport equations for the turbulent kinetic energy and energy dissipation rate are expressed in Eqs. (4) and (5), respectively:

$$\frac{\partial}{\partial t} (\rho \kappa) + \frac{\partial}{\partial x_i} (\rho V_i \kappa) = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_\kappa} \right) \frac{\partial \kappa}{\partial x_i} \right] + P_\kappa - \rho \epsilon + P_{\kappa b} \quad (4)$$

$$\frac{\partial}{\partial t} (\rho \epsilon) + \frac{\partial}{\partial x_i} (\rho V_i \epsilon) = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_i} \right] + \frac{\epsilon}{\kappa} (C_{\epsilon 1} P_\kappa - C_{\epsilon 2} \rho \epsilon + C_{\epsilon 1} P_{\kappa b}) \quad (5)$$

where

$$\mu_{eff} = \mu + \mu_t \quad (6)$$

$$\mu_t = C_\mu \rho \frac{k^2}{\epsilon} \quad (7)$$

$$P_\kappa = \mu_t \left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i} \right) \frac{\partial V_i}{\partial x_j} \quad (8)$$

$$P_{\kappa b} = -\frac{\mu_t}{\rho \sigma_\rho} g_i \frac{\partial \rho}{\partial x_i} \quad (9)$$

According to the dynamic renormalization group theory, the constants used in the κ – ϵ turbulence model are given as $C_\mu = 0.09$, $C_{\epsilon 1} = 1.44$, $C_{\epsilon 2} = 1.92$, $\sigma_\kappa = 1.0$, $\sigma_\epsilon = 1.3$, and $\sigma_\rho = 1.0$ [212].

The average temperature of the air was set at the arithmetic average of the grid faces. The infiltration heat load can be calculated from the heat energy difference of the air domain between the present and door-closed states. The infiltration heat load equation is given by Eq. (10).

$$Q = c_{air} m_{air} (T_{pre} - T_{ini}) \quad (10)$$

2.3. Boundary conditions

Fig. 3 shows schematics of the boundary conditions. As shown in Fig. 3(a), the air inside the truck body is sucked by fans, enters the cooling unit, is cooled while passing through the fin-tube heat exchanger, and is then released from the cooling unit. According to the fan manufacturer, the static pressure of the fan was 68 Pa, and the fan airflow was 827 m³/h. Before the transient simulation, a steady state simulation with the doors closed was conducted to set the initial condition. Since the steady state with no heat loss through the refrigerated truck wall with closed door was assumed before the transient simulation, therefore, the initial air temperature was set to a constant temperature of 253 K which was the same temperature as the fin-tube heat exchanger surface temperature based on the equilibrium state. The surface temperatures of the evaporator of the refrigeration cycle can be considered constant during the short unloading process of 40 s in this study. Also, steady state was assumed before conducting the transient analysis, the change in surface temperature of cooling coil can be considered as negligible during 40 s. Fig. 3(b) shows the boundary conditions of the truck body during the transient simulation. In cargo loaded cases, the results of the steady state simulation with the doors closed before opening the door were used to set the temperature and velocity boundary conditions for the transient simulation. For door opening, an open boundary condition was applied to the doors, and the opening temperature was set to 303 K, which is the typical outdoor temperature during summer. Applying opening boundary conditions to

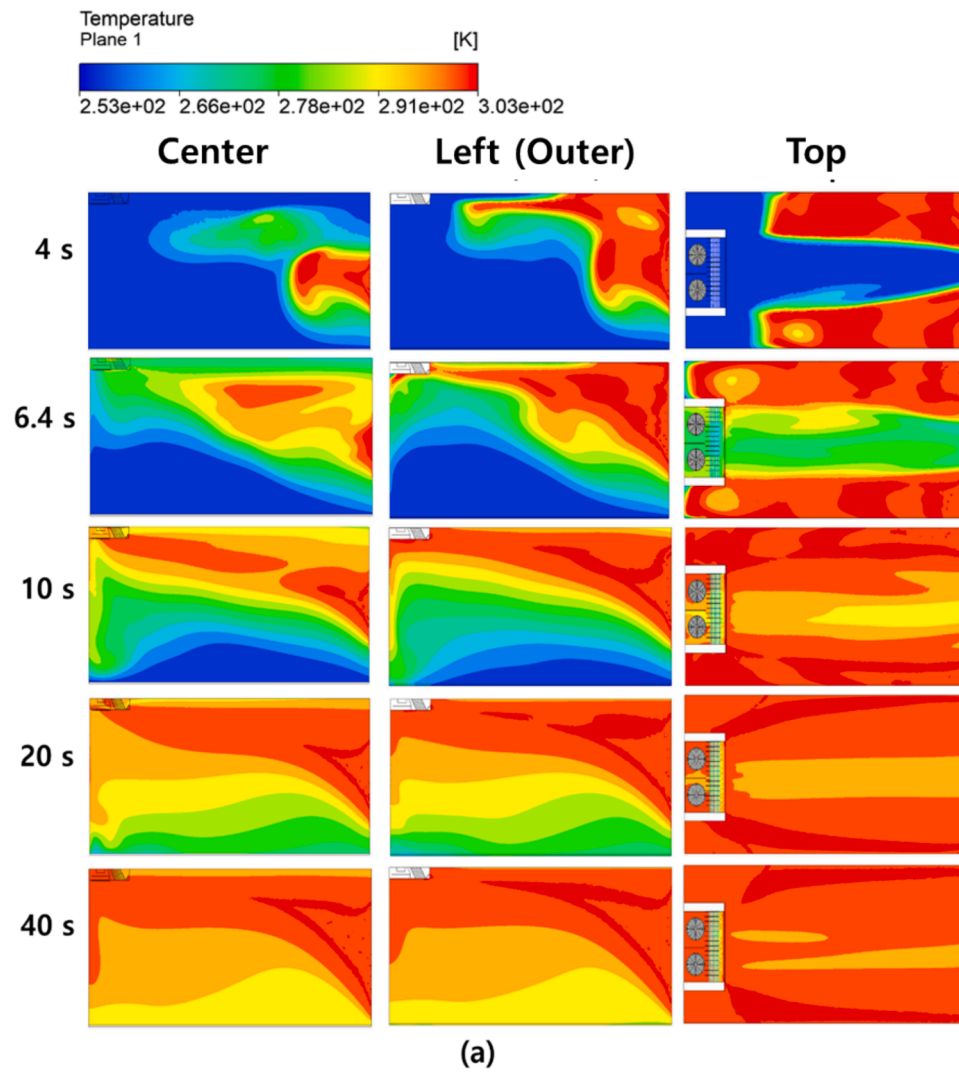


Fig. 7. (a) Temperature contour and (b) velocity vector of the air domain after opening all two doors of empty truck body.

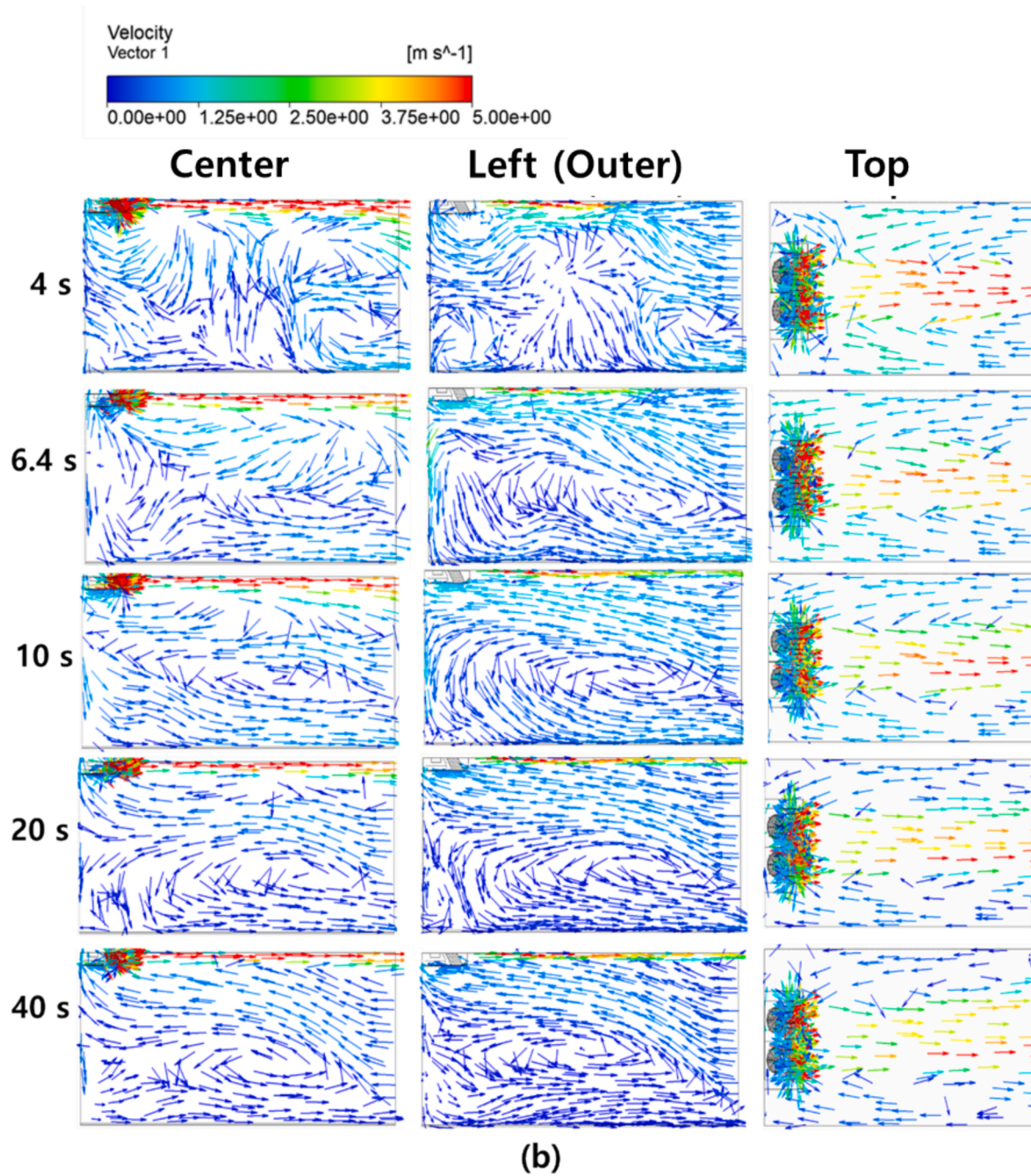


Fig. 7. (continued).

the opened door can be considered as reasonable because door opening motion is not taken into account; the effect of door opening motion on air flow is suppressed. Furthermore, the walls of the truck and cargo were considered adiabatic because the simulation time was short, and a normal insulation material was used [22,23]. The adiabatic boundary conditions at the truck wall and cargo walls can be expressed as Eq. (11).

$$\frac{\partial T}{\partial x_i} = 0 \quad (11)$$

The boundary conditions are summarized in Table 2.

2.4. Grid dependency test

Fig. 4 presents a schematic of the generated mesh, a detailed view of the section near the cooling unit, and the measurement locations for the grid-dependency test. As convective heat transfer occurs in the fin-tube heat exchanger, grids are predominantly generated near the cooling tubes. The grid-dependency test results are presented in Table 3. The grid size for the cooling unit was fixed 0.1 mm in the cooling unit region and 1.5 mm, 2 mm, 2.5 mm for the rest of the fluid domain for each model. The air temperatures and velocities at the three locations and the average air temperature and velocity of the truck body were considered as the parameters for the grid dependency index. The temperature at each location was calculated at four time intervals after opening both doors. Considering the computation time and precision, 17.23 million grids were accepted, presenting a maximum grid-dependency index of 2.88 % for the simulation in this study. The average skewness and the orthogonality of the accepted grid were 0.24 and 0.76, respectively, which satisfies the criteria of a good mesh: skewness under 0.25 and orthogonality above 0.7. The grid-dependency index equation is expressed in Eq. (12).

$$\theta = \frac{|i_{\text{finergrid}} - i_{\text{coarsergrid}}|}{i_{\text{finergrid}}} \times 100 \quad (12)$$

3. Experiments for validation

Experiments were conducted to validate the simulation results by comparing air temperatures inside the truck body. Fig. 5 shows a photograph of the experimental equipment and a schematic of the experiments. A commercial truck body capable of loading 2.5 tons of cargo was used in the experiment. The truck body dimensions were 2050 (width) × 2000 (height) × 4400 mm (depth). The air temperatures at 12 points inside the truck were measured every second using 12 wireless infrared (IR) temperature sensors. The sensor accuracy was ± 0.5 K and each sensor was installed 100 mm from the nearest wall. The fans were operated during the experiment, and the outside air temperature was set at 298 K. Initially, the doors were closed until the average air temperature inside the truck decreased to 263 K. Then, the right door was opened for 3 min. The experiment was conducted three times to verify the consistency of the experimental results. The measured data were corrected by considering the heat capacity of the wireless temperature sensor before comparison with the predicted data. The simulation was validated by comparing the predicted and experimental air temperatures at several different points and time steps after the right door was opened. As shown in Fig. 6, 92 % of the predicted data were within ± 3 K of the measured data error, and the predicted temperature well followed the trend of the measured temperature. Considering the characteristics of the transient simulation in which all errors are accumulated, the simulation results can be considered reliable.

4. Results and discussion

4.1. Two doors open without cargo

The temperature contours and velocity vectors of the air inside the

empty truck body after opening the two doors are shown in Fig. 7(a) and 7(b), respectively. The upper part of the truck body was highly influenced by the forced flow of the fans in the cooling unit. Therefore, the air flowed toward the exit of the truck at a high velocity, regardless of the time step. Because the cooling unit was in the middle of the truck body, the air velocity in the forced flow region was much higher at the center than that in the outer part of the truck body. Below the forced flow region, a natural convection region was formed by the buoyancy force caused by the density difference between the cold existing air inside the truck body and the hot outside air [10]. Relatively dense air flowed downward and was released outward along the lower part of the truck body. However, the outside air flowed into the upper part of the truck body owing to its low density. When the hot air reached the front side of the truck body, some of it was sucked up by the fans and entered the forced flow region, whereas the other air flowed downward owing to cold air leakage. The overall air temperature continuously increased as the cold and hot air mixed until the inside air temperature reached the outside temperature. Compared with the outer cross-sectional plane of the truck body, the air temperature increase was slower in the center cross-sectional plane of the truck body. This is because the center cross-sectional plane is more influenced by the air released from the cooling

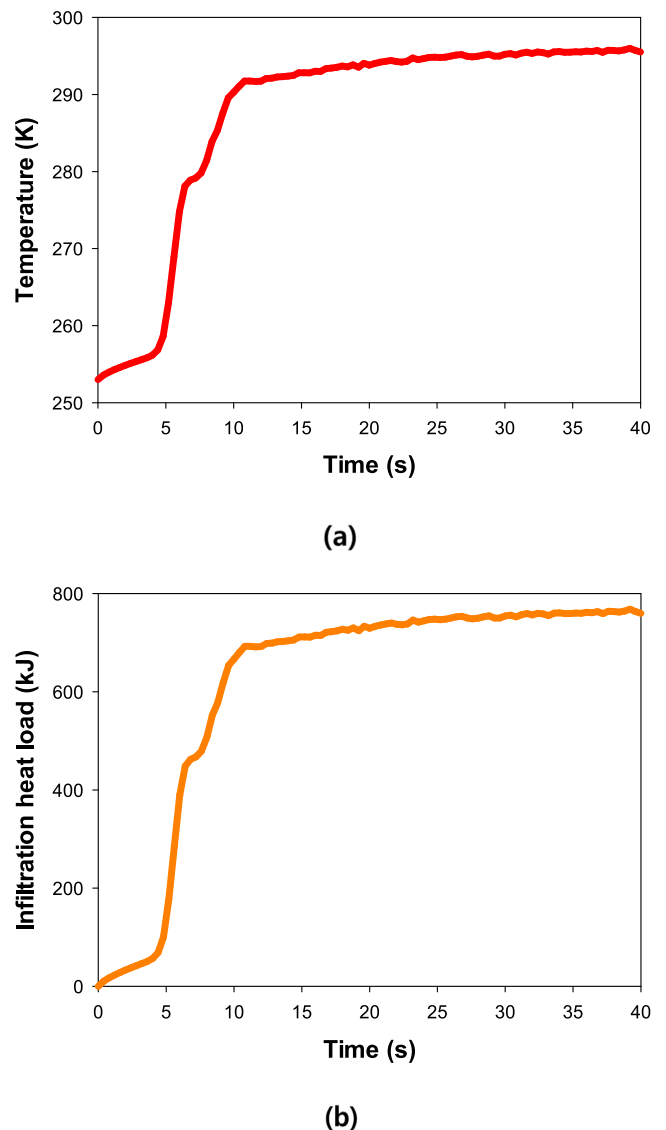


Fig. 8. (a) Average air temperature and (b) infiltration heat load change during transient simulation (empty truck body, two doors open, fan on).

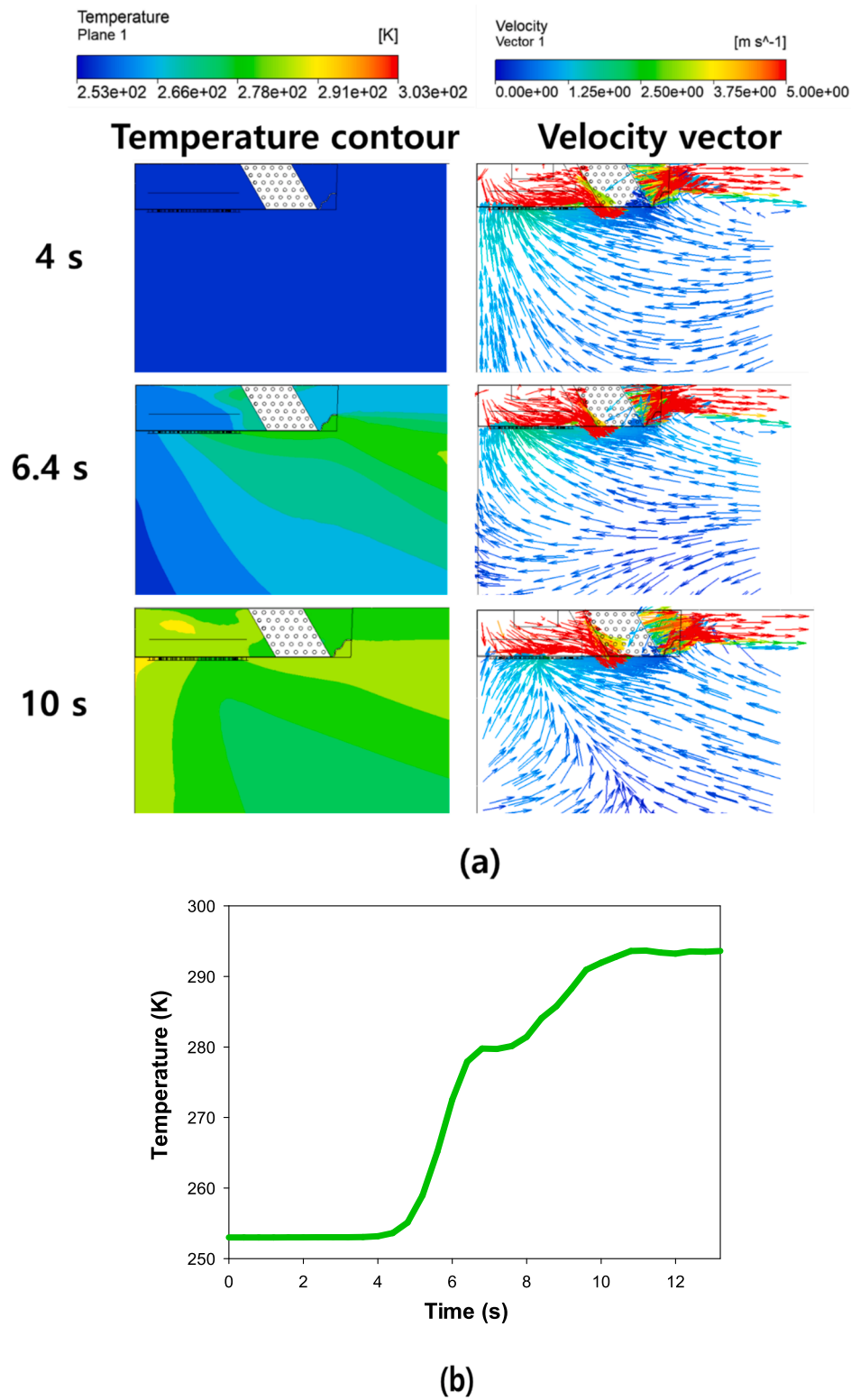


Fig. 9. (a) Temperature contour, velocity vector, and (b) air temperature at the cooling unit exit during the rapid increase of average air temperature.

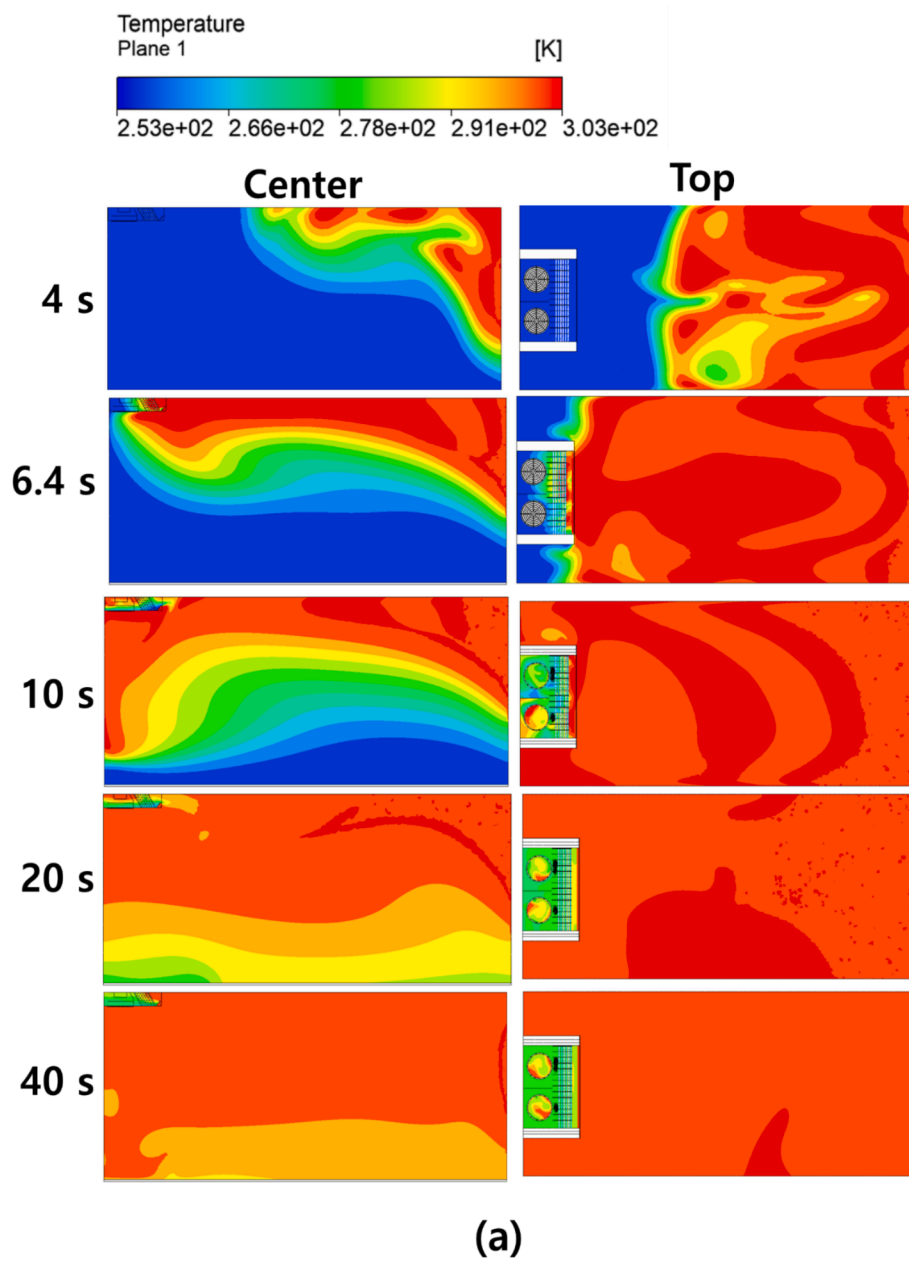


Fig. 10. (a) Temperature contour and (b) velocity vector of the air domain after opening both doors of an empty truck body and turning off the fans.

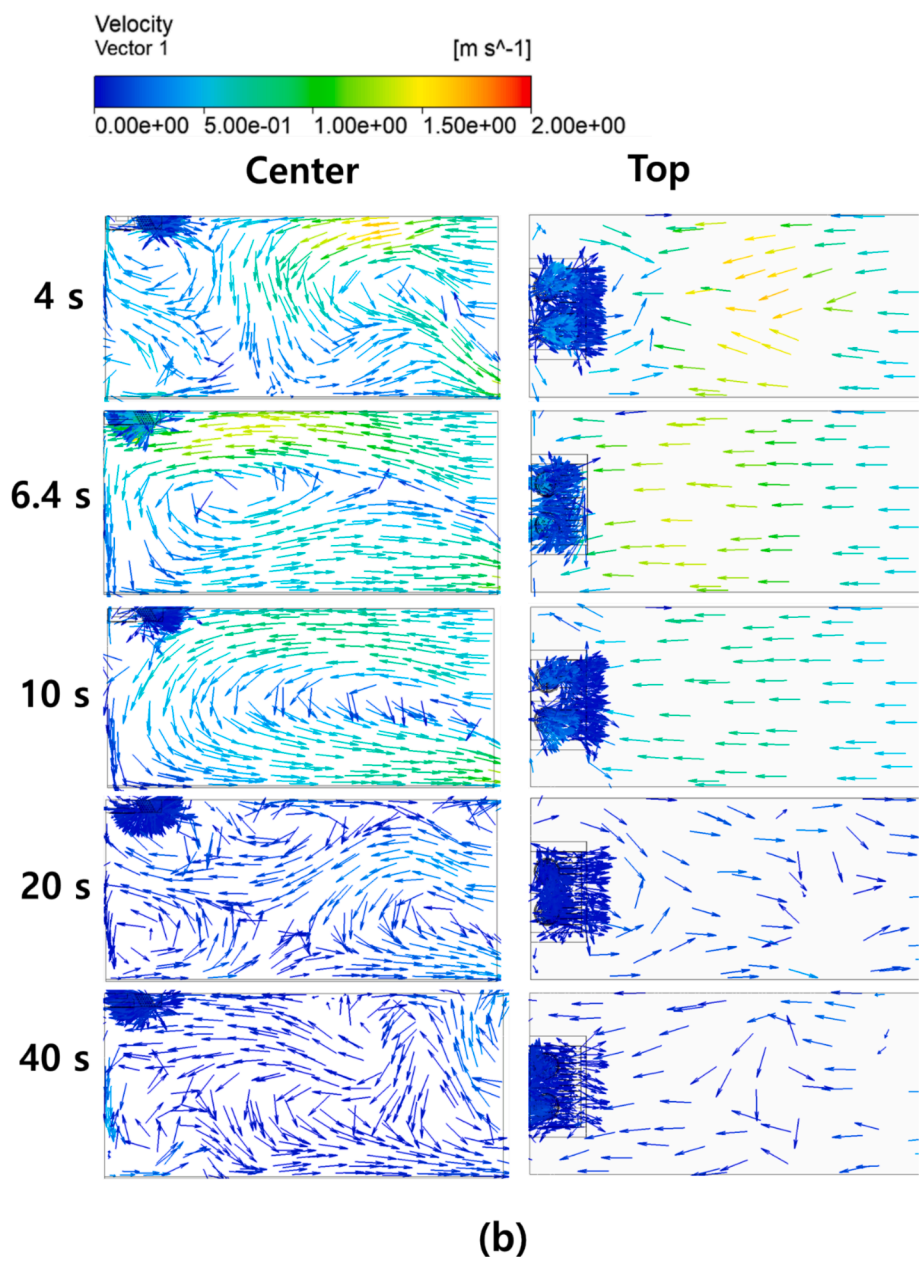


Fig. 10. (continued).

unit than the outer cross-sectional plane. The air released from the cooling unit was cooler than the surrounding air because of heat exchange in the fin-tube heat exchanger. Furthermore, the high air velocity blocked the inflow of outside air. Consequently, the overall temperature was lower in the center cross-sectional plane than in the outer cross-sectional plane at the same time step. Since the wall was in adiabatic boundary condition, all the temperature contours were perpendicular to walls in detail view.

Figs. 8(a) and 8(b) depict the temperature and infiltration heat load changes from 0 to 40 s. In the first 5 s, the air temperature slowly increased from 253 to 260 K. However, in the next 5 s, the air temperature rapidly increased from 260 to 290 K; this trend is presented in detail in Fig. 9. After a sharp increase, the increment rate of the air temperature decreased and the air temperature slowly increased to 295 K, which was close to the outside temperature of 303 K, until the end of the simulation. The infiltration heat load was increased from 0 to 740 kJ over 40 s. The tendency of the increase was consistent with the temperature change because the infiltration heat load was proportional to the temperature difference between the present and initial states.

Fig. 9(a) shows the temperature contour and velocity vector near the cooling unit, and Fig. 9(b) depicts the temperature at the cooling unit exit before, during, and after the rapid rise in the air temperature and infiltration heat load. As shown in Fig. 9(a) and Fig. 9(b), the influence of the outside air reached the cooling unit between $t = 4$ s and $t = 6.4$ s. Furthermore, in Fig. 9(c), the average air temperature of the cooling unit outlet surged from 253 K to 293 K from $t = 4$ s to $t = 10$ s. These results indicate that, after the outside air reaches the cooling unit, the cooling unit no longer discharges cold air to cool the truck body. Instead, it sucks up and discharges hot outside air, significantly increasing the average air temperature inside the truck body. After a rapid increase, the reduced temperature difference between the cold and hot regions reduces the mixing and buoyancy forces and suppresses the increment rate.

4.2. Effect of fan operation

Fig. 10 shows the temperature contours and velocity vectors of the air inside the empty truck body after opening the two doors and turning off the fans in the cooling unit. As shown in Fig. 10(a) and 10(b), there was no forced-flow region because the fans were not operating. Therefore, the flow inside the truck body was completely driven by the buoyancy force caused by the density difference between the existing cold air and the hot outside air. The overall air flow pattern well matches with the simulation results of the study of air infiltration pattern by De Micheaux et al. [13]. The flow momentum was active at the beginning; however, it decreased as the buoyancy force decreased, owing to the reduced density difference. At $t = 20$ s, the average air velocity inside the truck body was almost zero, indicating that the flow inside the truck body had almost reached a steady state. Compared with the fan-on case, the inflow of outside air was not blocked by the air blown from the cooling unit. Therefore, the air temperature of the top cross-sectional plane of the truck body reached the outside temperature faster than in the fan-on case. Especially, at around $t = 6.4$ s, inflow of outside air entering through upper part of the truck body reached the front side of the truck body and began to flow downward, promoting counterclockwise flow; this counterclockwise flow further accelerated outflow of the existing cold air in the bottom part of the truck body.

Fig. 11 shows a comparison of the air temperature and infiltration heat load changes over time when the fans are on and off. During the first few seconds, the air temperature increased slowly and then changed rapidly. This is due to the promoted outflow of air at approximately $t = 6.4$ s as shown in Fig. 10. However, unlike the fan-on case, the temperature dropped to 276 K after a sharp rise in the fan-off case; this trend is detailed in Fig. 12. The air temperature then increased slowly and approached the outdoor temperature. The air temperature did not increase above 290 K until the end of the simulation in the fan-off case,

whereas it exceeded 290 K in the fan-on case. These results indicate that the activating the cooling unit suppresses the infiltration heat load to increase at first, but eventually results in a higher infiltration heat load. This means that turning off the cooling unit right after opening the doors is desirable.

Fig. 12(a) shows the mass flow rate of the outside air inflow at the truck exit. When the fans were not operating, the mass flow rate was relatively low in the first few seconds, below 1.2 kg s^{-1} . This is because the fans were operating before the door opened, and the remaining swirling flow near the truck exit blocked the inflow of the outside air. However, the inflow of the outside air increased and reached a peak at $t = 6$ s as the remaining swirling flow diminished. Considering that the mass flow rate increased from $t = 4.8$ s to $t = 7.2$ s, the temperature surge was due to the increased inflow of outside air after the effect of the remaining swirling flow diminished. Compared with the fan-on case, the mass flow rate in the fan-off case was lower during the entire simulation because of the strong forced flow from the cooling unit toward the outside. Especially, after $t = 10$ s, the mass flow rate in the fan-off case gradually decreased and got close to zero while the mass flow rate in the fan-on case maintained a value above 1 kg s^{-1} owing to the outward flow by the fans. The air temperature decrease, shown between $t = 5.2$ s and $t = 7.2$ s, in the fan-off case was due to the heat exchange in the fin-

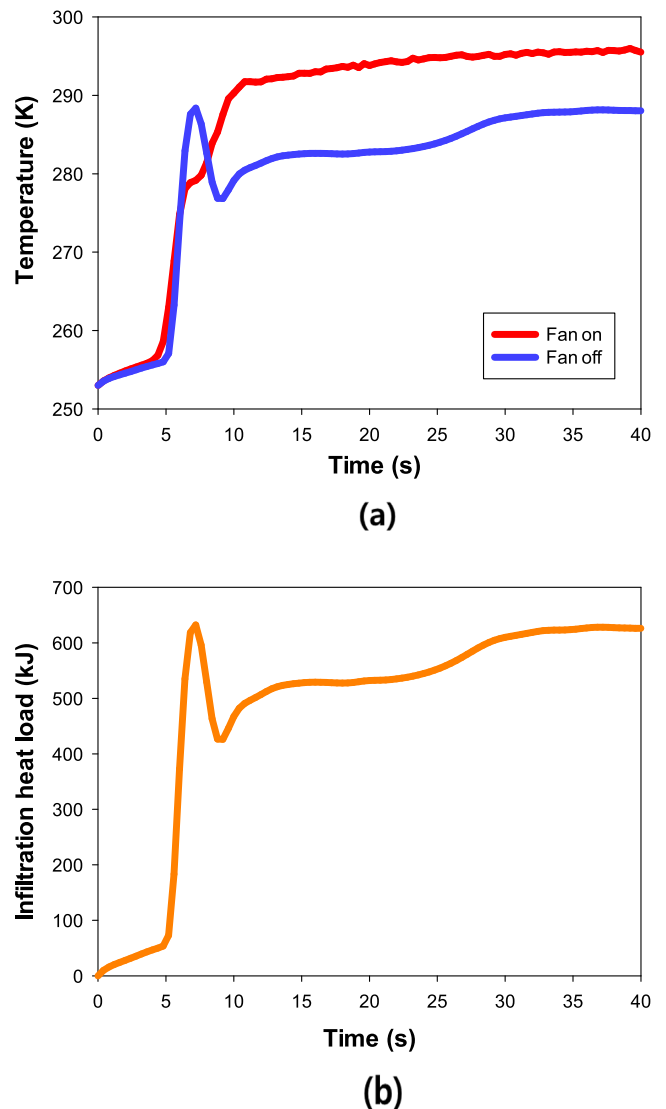


Fig. 11. (a) Average air temperature comparison (fan on and off) and (b) infiltration heat load change after turning off the fans.

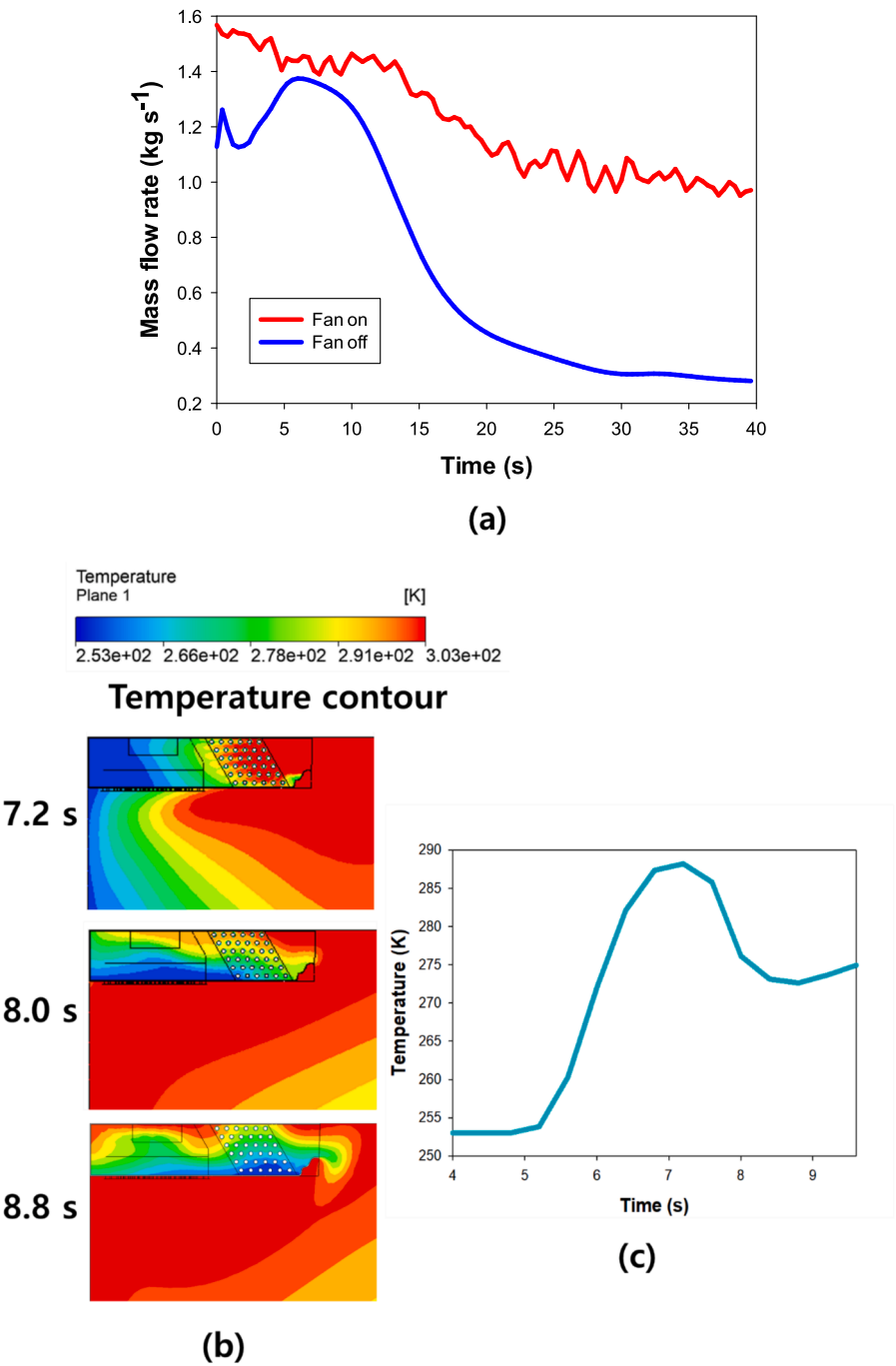


Fig. 12. (a) Comparison of the mass flow rate at the truck exit, (b) temperature contour, and (c) air temperature change near the tubes of the fin-tube heat exchanger.

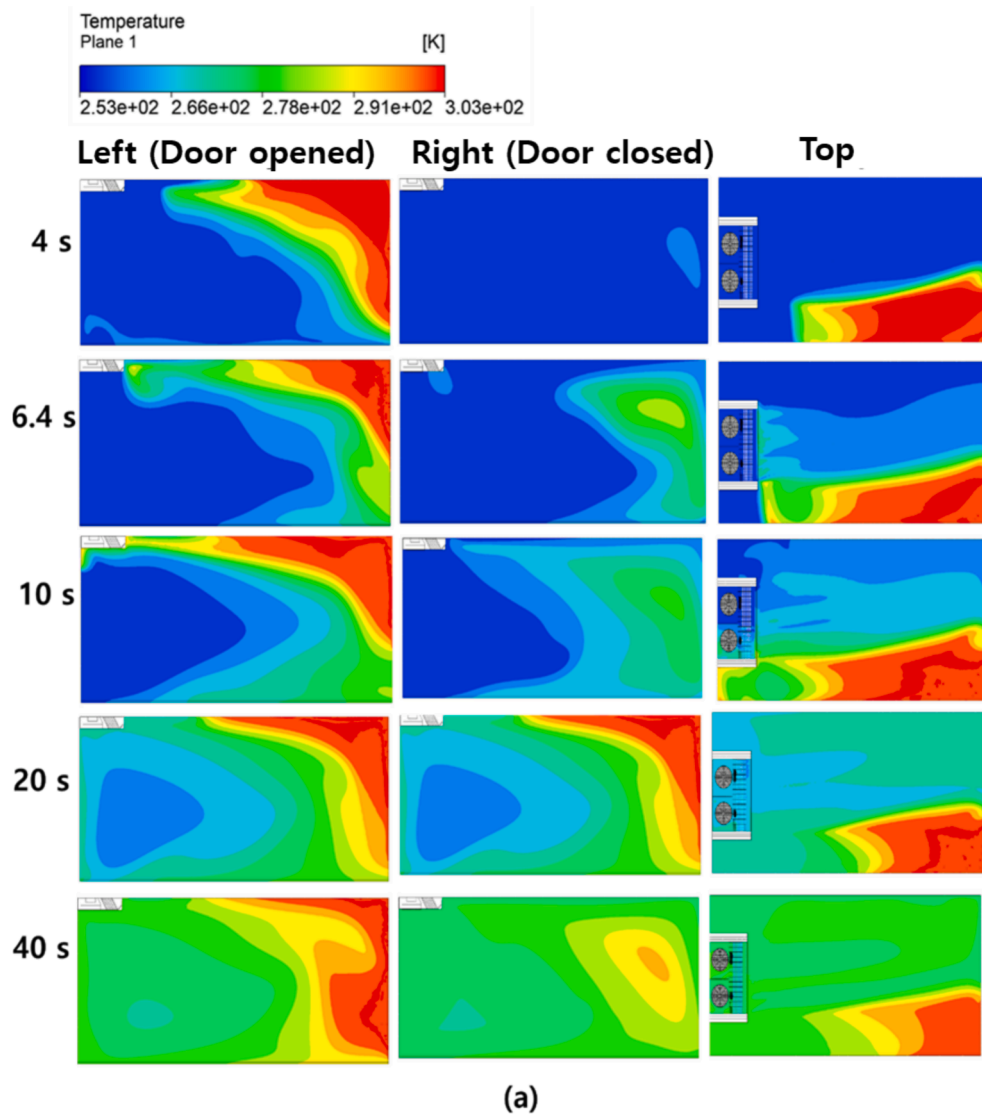


Fig. 13. (a) Temperature contour and (b) velocity vector of the air domain after opening only the left door of an empty truck body.

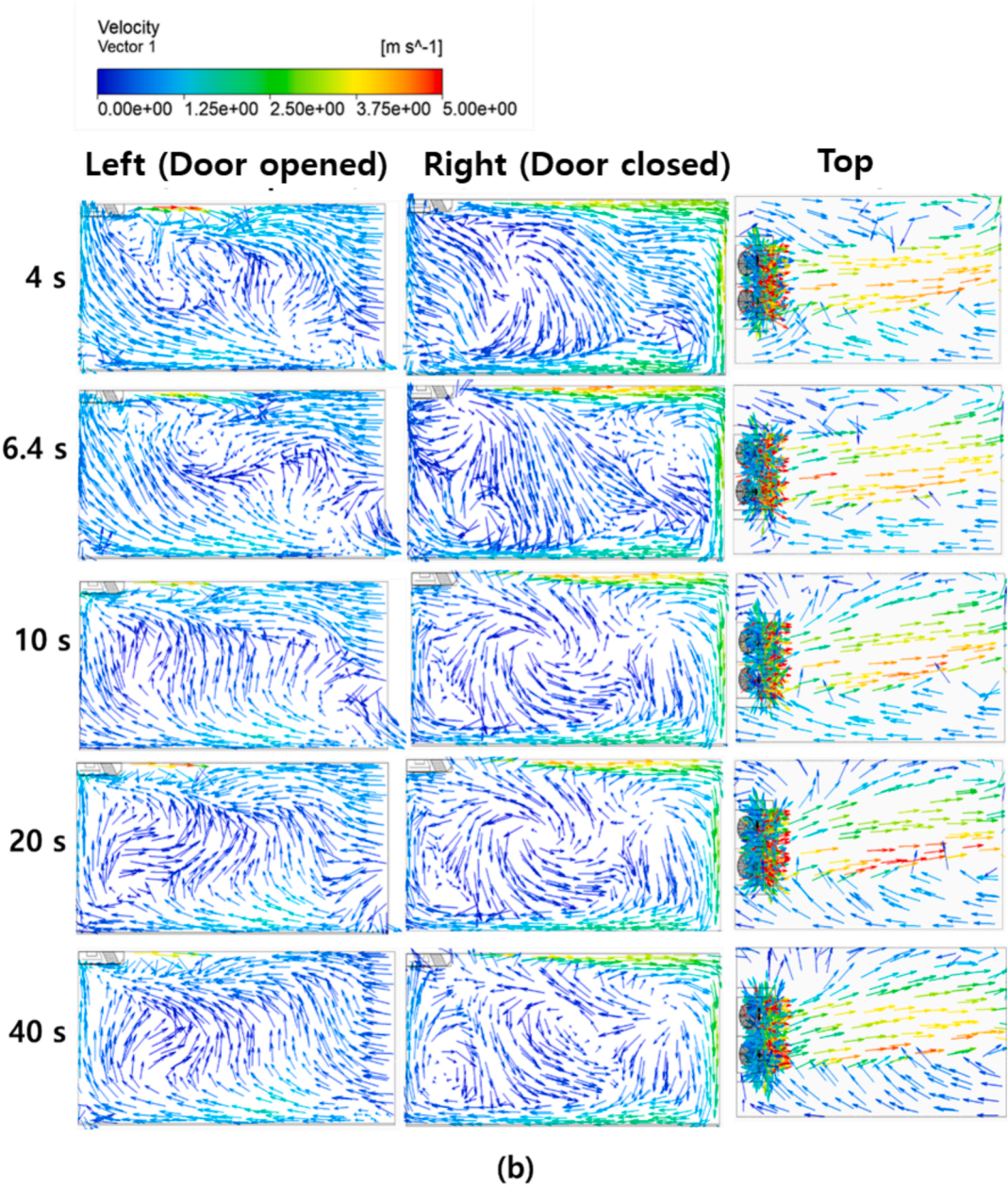


Fig. 13. (continued).

tube heat exchanger. Fig. 12(b) and 12(c) depict the flow near the cooling unit and air temperature change near the fin-tube heat exchanger, respectively. Because the tubes maintained a constant temperature of 253 K and the fans were off, the hot outside air that entered the cooling unit was cooled, and this caused the average air temperature to decrease. In other words, the cooled outside air that passed through the fin-tube heat exchanger flowed toward the outside in the fan-on case, resulting in a higher average temperature, whereas it flowed downward inside in the fan-off case, resulting in a lower average temperature.

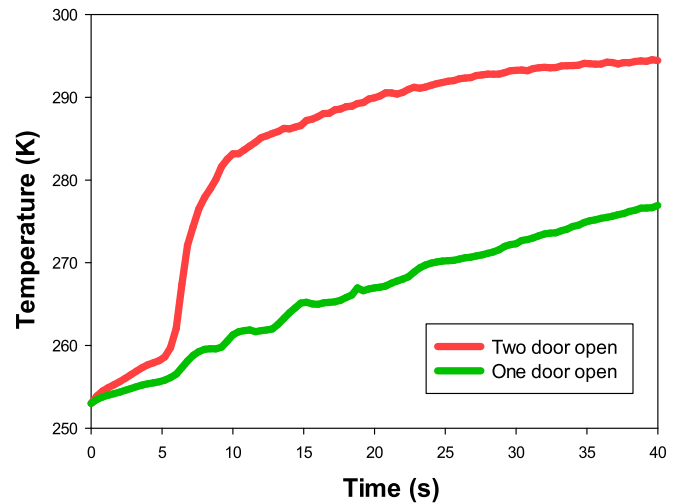
4.3. Effect of one open door

Fig. 13 depicts the temperature contour and velocity vector of the air inside the truck body after opening only the left door. On the left side of the truck body, the flow pattern and temperature distribution were similar to those observed when the two doors were opened. A forced-flow region was dominant on the top side of the truck body, and a natural convection region was observed below the forced-flow region. However, in the right cross-sectional plane of the truck body, the flow pattern differs because of the closed door. Because the closed door acted as a wall, the inflow of outside air and leakage of the remaining cold air were blocked. Furthermore, the air blown from the cooling unit circulated along the walls instead of exiting the truck from the top side of the truck body. In other words, the airflow in the right cross-sectional plane of the truck body was similar to that before the door was closed. In addition, the high air velocity of the circulation suppressed the influence of the inflow of hot outside air from the open door. Therefore, the cold air circulation and blockage of the outside air inflow at the closed door maintained a cold temperature on the right side of the truck body.

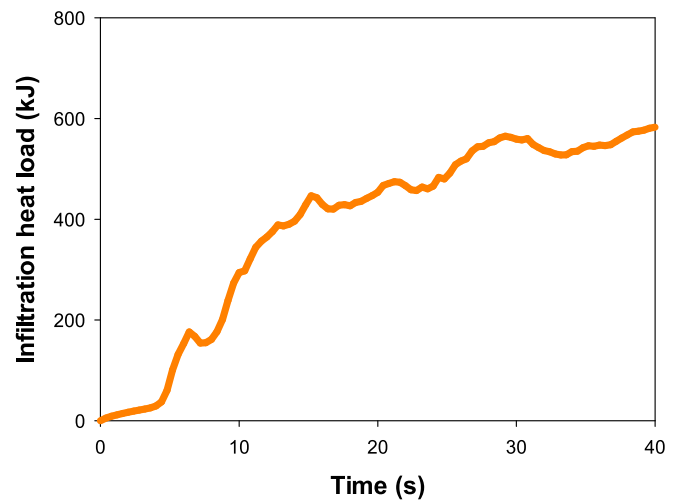
Fig. 14 presents a comparison of the air temperature and infiltration heat load changes over time between opening one door and opening both doors. The average air temperature was lower when one door was opened during the simulation. Unlike the case when both doors were opened, a dramatic increase in air temperature was not observed, and the temperature rose 0.6 – 1.0 K per second steadily during the simulation. This result indicates that the cold air circulation and outside air blockage caused by the closed door prevent the rapid rise in air temperature by allowing less outside air to be sucked into the cooling unit. Furthermore, because half of the truck body remained cold, the air temperature increase was slower than when both doors were opened. The infiltration heat load increased slowly, reaching 580 kJ at the end of the simulation. The decrease of rate of air temperature increase compared to both doors opened case well matches to the study conducted by Taher et. al [11]. The overall temperature was also lower than the fan-off case, which indicates that the forced convection of the cooling unit has an effect on reducing the infiltration load if wall exists in the truck exit.

4.4. Effect of cargo loading

Fig. 15 shows the temperature contour and velocity vector of the air inside the truck body after both doors were opened when the truck body was half-loaded. Most of the flow patterns were similar to those observed when the truck body was empty. However, because the cargo acted as a wall, a flow pattern that occurred near the cooling unit and front side of the empty truck body appeared after the outside air reached the back side of the cargo. When the outside air reached the cargo at approximately $t = 6.4$ s, some of it flowed upward, entered the forced-flow region, and was discharged through the truck exit. However, some flowed downward and was released outward along the lower part of the truck body, and the rest flowed through the narrow space between the truck walls and the cargo. The temperature and velocity distributions were relatively irregular compared to those of the empty truck body because of the larger turbulent region caused by the narrow space in the front part of the truck body.



(a)



(b)

Fig. 14. (a) Average air temperature comparison (two door open/one door open) and (b) infiltration heat load change after opening only one door.

Fig. 16 shows the temperature contour and velocity vector of the air inside the truck body when the truck body was fully loaded after both doors were opened. Because most of the space inside the truck body was full of cargo, hot outside air flowed through the narrow gap between the walls of the cargo and truck body. Therefore, the airflow inside the truck body was similar to that inside a hollow rectangular pipe. In the upper part of the truck body, the air temperature was lower than at other locations owing to the effect of the cooling unit. Furthermore, the air temperature in the upper part was lower than that in the other two-door open cases at the same time step. This is because the cargo blocks the inflow of the outside air from the inlet, whereas the influence of the forced flow is still dominant. However, the lower part of the truck body quickly approached the outside air temperature, unlike in the other cases, because the outside air flowed along a narrow bottom path. The overall temperature and velocity distributions showed a more irregular pattern than those in the half-loaded case because of the larger turbulent region.

Fig. 17 shows a comparison of the air temperature and infiltration heat load changes over time for different loading conditions. When the truck body was half-loaded, the overall temperature increase pattern was similar to the case with an empty truck body because the flow

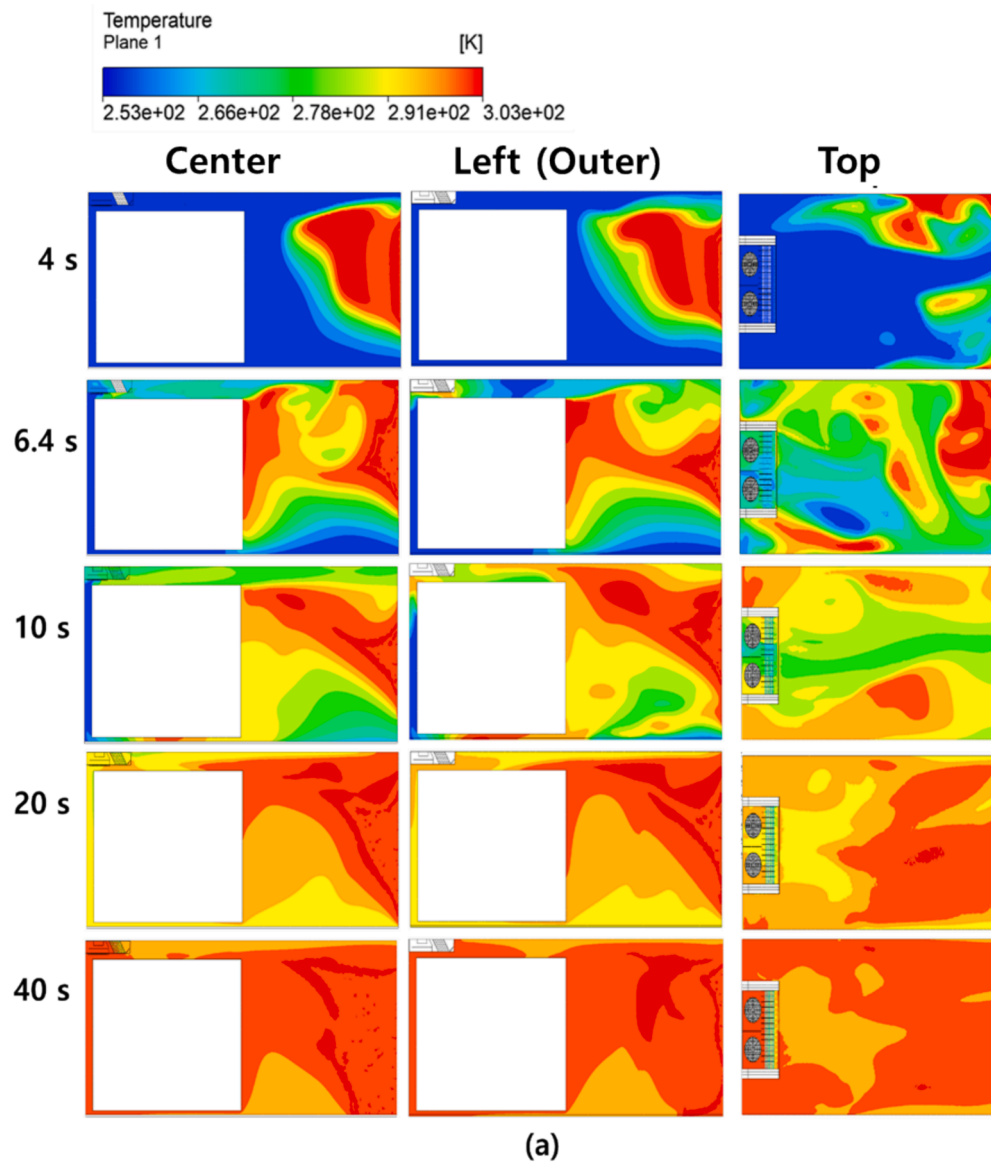


Fig. 15. (a) Temperature contour and (b) velocity vector of the air domain after opening all two doors when the truck body is half loaded.

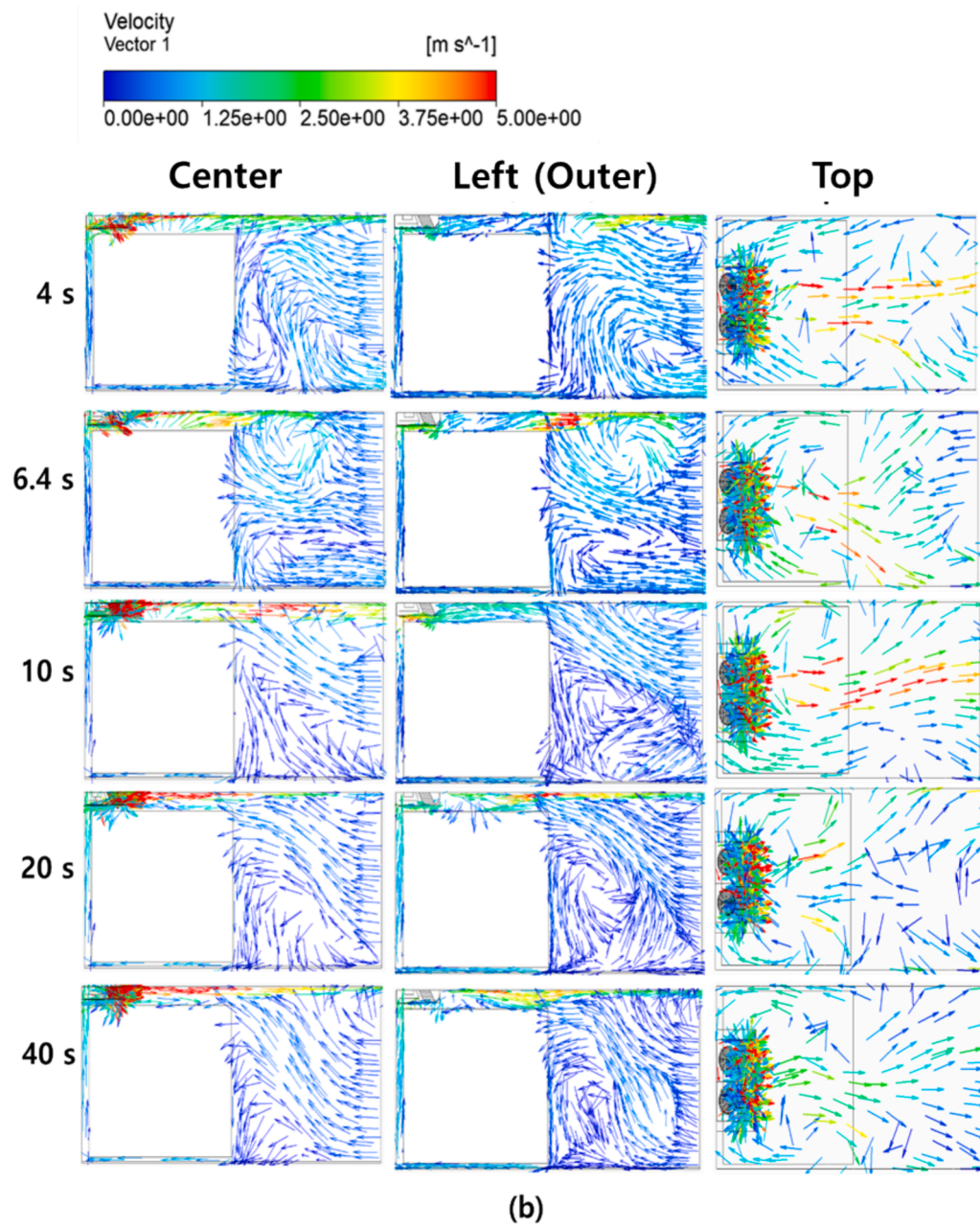


Fig. 15. (continued).

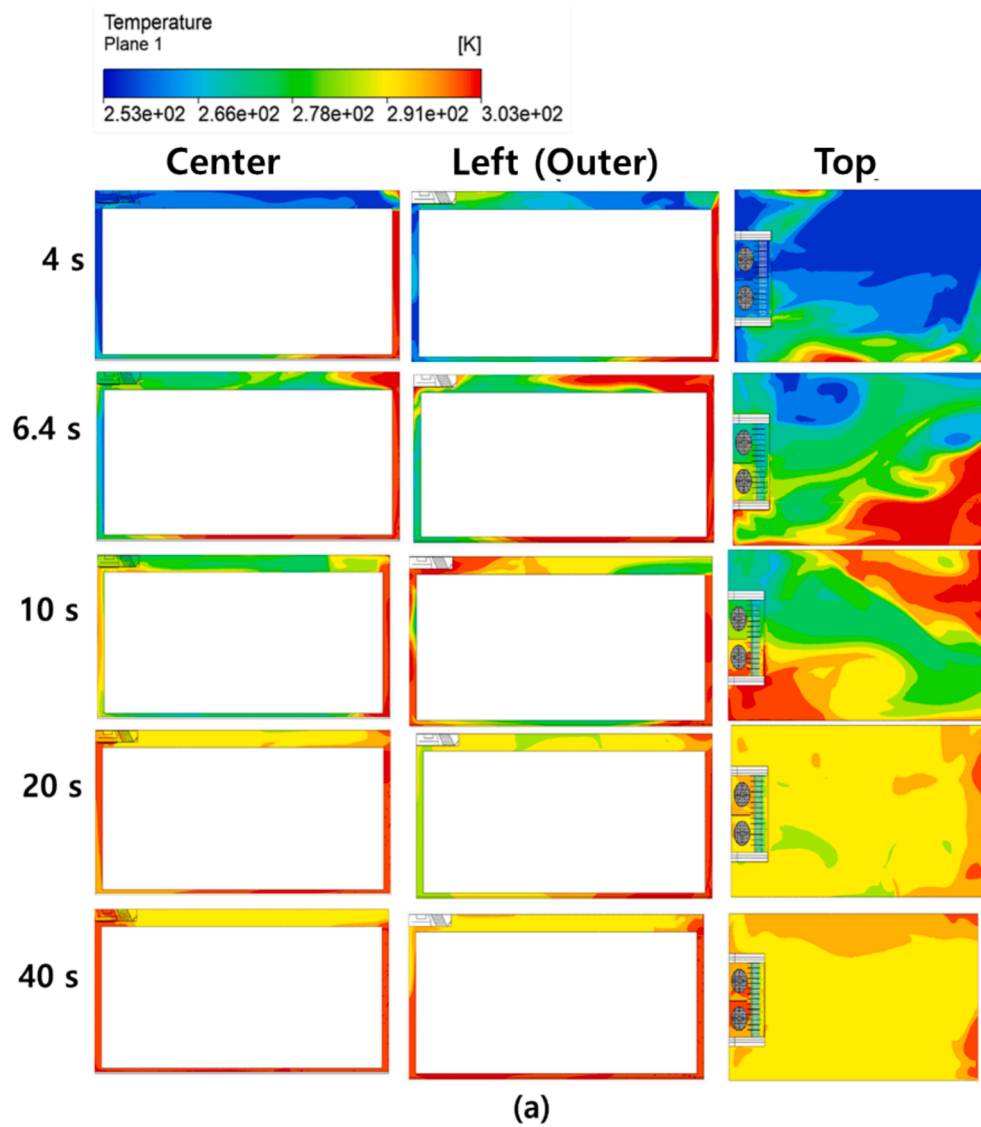


Fig. 16. (a) Temperature contour and (b) velocity vector of the air domain after opening both doors when the truck body is fully loaded.

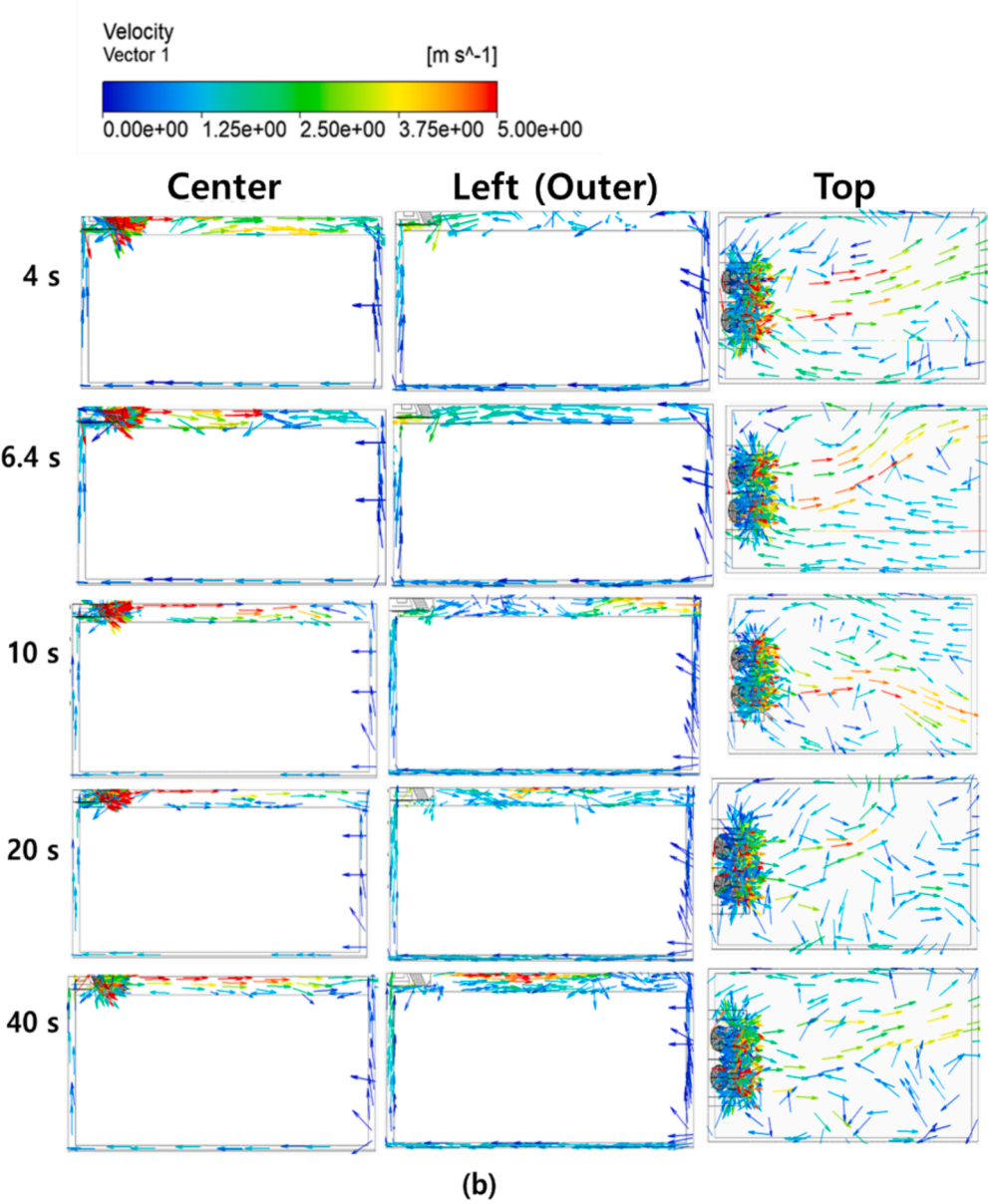


Fig. 16. (continued).

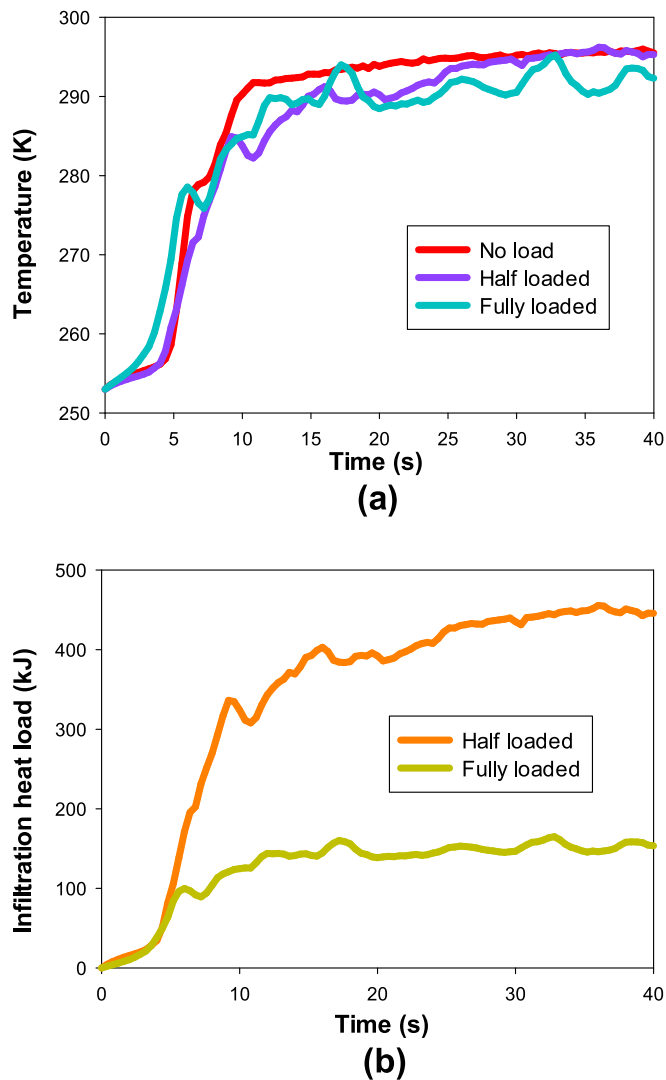


Fig. 17. (a) Average air temperature comparison (empty/half-loaded/fully loaded) and (b) infiltration heat load change when cargo is loaded.

pattern beyond the cargo was almost the same for the two cases. However, the temperature increase was relatively slower for the half-loaded case than that with no load; the air temperature increased from 263 K to 284 K between $t = 5.2$ s and $t = 10$ s, which was approximately 23 % slower than with no load. This was because the cargo suppressed the outside air from reaching the front side, preventing the existing cold air from flowing outward; only approximately 40 % of the total outside air inflow flowed toward the front side of the truck body through the gap between the cargo and the walls. Furthermore, outside air could not directly reach the cooling unit. Therefore, the front half of the truck body, particularly near the cooling unit, remained cooler than the empty truck body at the same time step. The overall pattern of the increase was similar to that in the fully loaded case. However, a sharp rise in the air temperature occurred earlier in the case with the fully loaded truck body than in the other cases because the mass of air inside the fully loaded truck body was only 21.9 % and 37.0 % of the empty and half-loaded truck bodies, respectively. Furthermore, the increasing pattern was relatively irregular, because the truck body resembled a completely turbulent pipe with a Reynolds number above 1×10^4 . The infiltration heat loads in the half-loaded and fully loaded cases reached 445 and 153 kJ at the end of the simulation, respectively, which were lower than those of the empty truck body owing to the air mass difference. The delay on the rapid temperature increase when cargo is loaded indicates

that proper cargo allocation can be a method to reduce infiltration heat load.

5. Conclusions

- No load, two doors open, fan-on: A forced flow region was developed in the upper part of the truck body because of the fans inside the cooling unit. Below the forced-flow region, a natural-convection region caused by the density difference was observed. A sharp increase (6 K s^{-1}) in the air temperature occurred because the cooling unit started to suck hot outside air instead of cold inside air, thereby increasing the inflow air flow rate.
- Effect of fan-off case: The flow was completely driven by the buoyancy force in the fan-off case. A counterclockwise flow trend was promoted. The air temperature increased rapidly at first to 289 K at the first 7 s because the increased inflow of outside air after the outside air reached the front side of the truck body, which was similar to the fan-on case. However, after the rapid increase, the average air temperature decreased to 276 K for the next 2 s, owing to the heat exchange of the hot inflow of outside air with the fin-tube heat exchanger.
- Effect of one door closed: Although the left side effect was similar to when both doors were opened, the air blown from the cooling unit circulated along the walls instead of exiting the truck on the closed right side. The average air temperature remained around 20 – 30 K lower than when both doors were opened during the simulation. Owing to the cold air circulation and outside air blockage, a dramatic increase in air temperature was not observed, and the temperature rose $0.6 - 1.0 \text{ K s}^{-1}$ steadily during the simulation.
- Effect of cargo loading: The overall tendencies of the flow pattern, temperature, and infiltration for half-loaded and fully loaded were similar to those in the case with no load. However, the rate of temperature increase was approximately 23 % slower for the half-loaded case than for the no-load case because the cargo prevented the outside air from directly reaching the front side of the truck body. Conversely, the air temperature increase was fastest in the case of full loading, compared with the other cases because of the small air mass.

The results of this study show the air flow and heat transfer behavior in the transient state of the truck body after opening the door. This study contains detailed investigation on the airflow and rapid change in air temperature in several different locations at numerous timesteps. In response to the global trend of electrification, even in refrigerated trucks, the result of this study can be used to optimize the operating strategy and enhance the geometry of the refrigerated truck by minimizing the factors that lead high and rapid heat loss, ultimately improving energy efficiency.

CRediT authorship contribution statement

Hyun Sung Hwang: Methodology, Investigation, Writing – original draft, Visualization. **Youngho Rhee:** Formal analysis, Validation. **Dongchan Lee:** Supervision, Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry, and Energy (MOTIE) of the Republic of Korea (No. 20212020800090).

Data availability

The authors are unable or have chosen not to specify which data has been used.

References

- [1] C. James, Food transportation and refrigeration technologies—Design and optimization, *Sustainable Food Supply Chains: Planning, Design, and Control through Interdisciplinary Methodologies* (2019) 185–199. Doi: 10.1016/B978-0-12-813411-5.00013-2.
- [2] S. Yun, J. Kwon, W. Cho, D.C. Lee, Y. Kim, Performance improvement of hot stamping die for patchwork blank using mixed cooling channel designs with straight and conformal channels, *Appl. Therm. Eng.* 165 (2020), <https://doi.org/10.1016/j.applthermaleng.2019.114562>.
- [3] R. Shashi, R. Cerchione, P. Singh, A. Centobelli, Shabani, Food cold chain management: From a structured literature review to a conceptual framework and research agenda, *Int. J. Logist. Manag.* 29 (2018) 792–821, <https://doi.org/10.1108/IJLM-01-2017-0007/FULL/PDF>.
- [4] A. Maiorino, F. Petruzzello, C. Aprea, Refrigerated transport: State of the art, technical issues, innovations and challenges for sustainability, *Energies (Basel)* 14 (2021), <https://doi.org/10.3390/en14217237>.
- [5] A. Alison Smith Paul Watkiss Geoff Tweddle Alan McKinnon Mike Browne Alistair Hunt Colin Treleven Chris Nash Sam Cross, P. Watkiss, Title The Validity of Food Miles as an Indicator of Sustainable Development: Final report, n.d.
- [6] J.B. Skjærseth, Linking EU climate and energy policies: Policy-making, implementation and reform, *Int. Environ. Agreem.* 16 (2016) 509–523, <https://doi.org/10.1007/S10784-014-9262-5/METRCS>.
- [7] S. Tian, Y. Gao, S. Shao, H. Xu, C. Tian, Development of an unsteady analytical model for predicting infiltration flow rate through the doorway of refrigerated rooms, *Appl. Therm. Eng.* 129 (2018) 179–186, <https://doi.org/10.1016/J.APPLTHERMALENG.2017.10.023>.
- [8] Y. Xie, J.K. Brecht, C.E. Abraham, E.R. Bornhorst, Y. Luo, A.L. Monge, K. Vorst, W. Brown, Improving temperature management and retaining quality of fresh-cut leafy greens by retrofitting open refrigerated retail display cases with doors, *J. Food Eng.* 292 (2021) 110271, <https://doi.org/10.1016/J.JFOODENG.2020.110271>.
- [9] S.A. Tassou, G. De-Lille, Y.T. Ge, Food transport refrigeration – Approaches to reduce energy consumption and environmental impacts of road transport, *Appl. Therm. Eng.* 29 (2009) 1467–1477, <https://doi.org/10.1016/J.APPLTHERMALENG.2008.06.027>.
- [10] A. Rai, S.A. Tassou, Environmental impacts of vapour compression and cryogenic transport refrigeration technologies for temperature controlled food distribution, *Energy Convers. Manag.* 150 (2017) 914–923, <https://doi.org/10.1016/J.ENCONMAN.2017.05.024>.
- [11] M.A. Ben Taher, M. Ahachad, M. Mahdaoui, Y. Zeraoui, T. Kousksou, A survey of computational and experimental studies on refrigerated trucks, *J. Energy Storage* 47 (2022) 103575, <https://doi.org/10.1016/J.EST.2021.103575>.
- [12] E.J. Sarabia Escrivá, V. Soto Francés, J.M. Pinazo Ojer, Comparison of annual cooling energy demand between conventional and inflatable dock door shelters for refrigerated and frozen food warehouses, *Therm. Sci. Eng. Prog.* 15 (2020) 100386, <https://doi.org/10.1016/J.TSEP.2019.100386>.
- [13] T. Lafaye De Micheaux, M. Ducoulombier, J. Moureh, V. Sartre, J. Bonjour, Experimental and numerical investigation of the infiltration heat load during the opening of a refrigerated truck body, *Int. J. Refrig.* 54 (2015) 170–189, <https://doi.org/10.1016/J.IJREFRIG.2015.02.009>.
- [14] M.A. Ben Taher, M. Mahdaoui, T. Kousksou, Y. Zeraoui, M. Ahachad, Numerical study of the aero-thermal performance for different scenarios of a refrigerated truck using URANS, *J. Clean. Prod.* 320 (2021) 128775, <https://doi.org/10.1016/j.jclepro.2021.128775>.
- [15] F. Moerman, K. Fikiin, J. Evans, A.M. Foster, Air infiltration control to reduce hygiene hazards in refrigerated food processing and storage facilities, in: *Hygienic Design of Food Factories*, Second Edition, Elsevier, 2023: pp. 319–378. Doi: 10.1016/B978-0-12-822618-6.00003-4.
- [16] C.P. Tso, S.C.M. Yu, H.J. Poh, P.G. Jolly, Experimental study on the heat and mass transfer characteristics in a refrigerated truck, *Int. J. Refrig.* 25 (2002) 340–350, [https://doi.org/10.1016/S0140-7007\(01\)00015-9](https://doi.org/10.1016/S0140-7007(01)00015-9).
- [17] A. Rai, J. Sun, S.A. Tassou, Numerical investigation of the protective mechanisms of air curtain in a refrigerated truck during door openings, *Energy Procedia* 161 (2019) 216–223, <https://doi.org/10.1016/J.EGYPRO.2019.02.084>.
- [18] M.A. Ben Taher, T. Kousksou, Y. Zeraoui, M. Ahachad, M. Mahdaoui, Thermal performance investigation of door opening and closing processes in a refrigerated truck equipped with different phase change materials, *J. Energy Storage* 42 (2021) 103097, <https://doi.org/10.1016/J.EST.2021.103097>.
- [19] J. Jeong, A.E. Benchikh Le Hocine, S. Croquer, S. Poncet, B. Michel, J. Bonjour, Numerical analysis of the thermo-aerodynamic behavior of air during the opening of the door of a refrigerated truck trailer equipped with cold plates, *Appl. Therm. Eng.* 206 (2022) 118057, <https://doi.org/10.1016/J.APPLTHERMALENG.2022.118057>.
- [20] J. Jeong, A.E. Benchikh, L. Hocine, S.D. Croquer, A.E. Benchikh Le Hocine, S. Croquer, S. Poncet, J. Bonjour, B. Michel, Numerical Simulation of the Heat Transfer in a Refrigerated Trailer Equipped with Eutectic Plates for Frozen Food Delivery Flow dynamics and heat transfer of ice slurry in heat exchangers View project Transcritical CO₂ ejectors in refrigeration View project Numerical Simulation of the Heat Transfer in a Refrigerated Trailer Equipped with Eutectic Plates for Frozen Food Delivery, 2021. <https://www.researchgate.net/publication/351914619>.
- [21] G.C. Papageorgakis, D.N. Assanis, Comparison of linear and nonlinear rng-based k-epsilon models for incompressible turbulent flows, *Numer. Heat Transf. B Fundam.* 35 (1999) 1–22, <https://doi.org/10.1080/104077999275983>.
- [22] M. Ahmed, O. Meade, M.A. Medina, Reducing heat transfer across the insulated walls of refrigerated truck trailers by the application of phase change materials, *Energy Convers. Manag.* 51 (2010) 383–392, <https://doi.org/10.1016/J.ENCONMAN.2009.09.003>.
- [23] A. Kan, W. Zhu, T. Wang, Y. Yuan, X. Zhang, Thermal performance assessment of cold chain chamber with vacuum insulation panel envelope layer, *Clean. Eng. Technol.* 4 (2021) 100157, <https://doi.org/10.1016/J.CLET.2021.100157>.